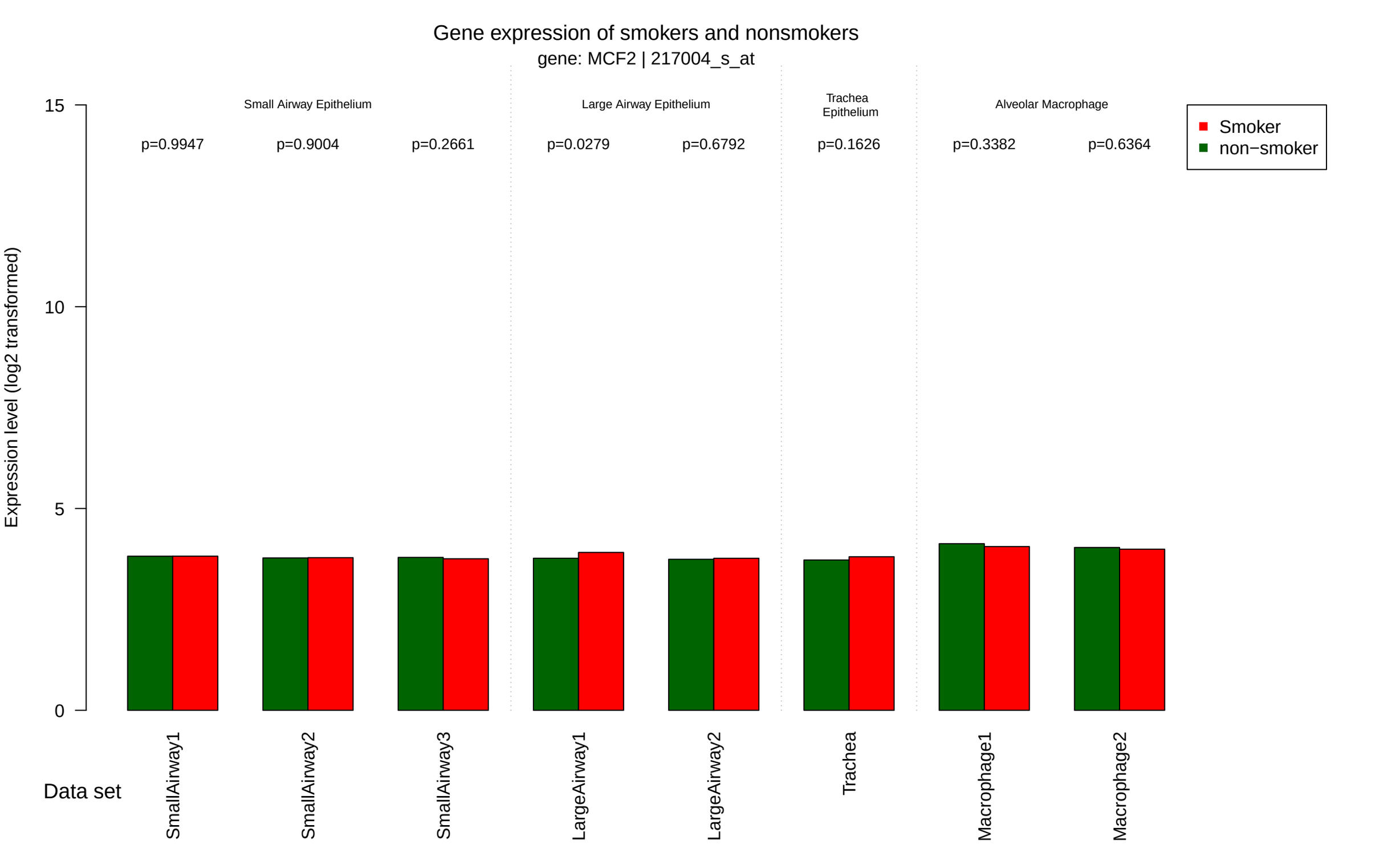
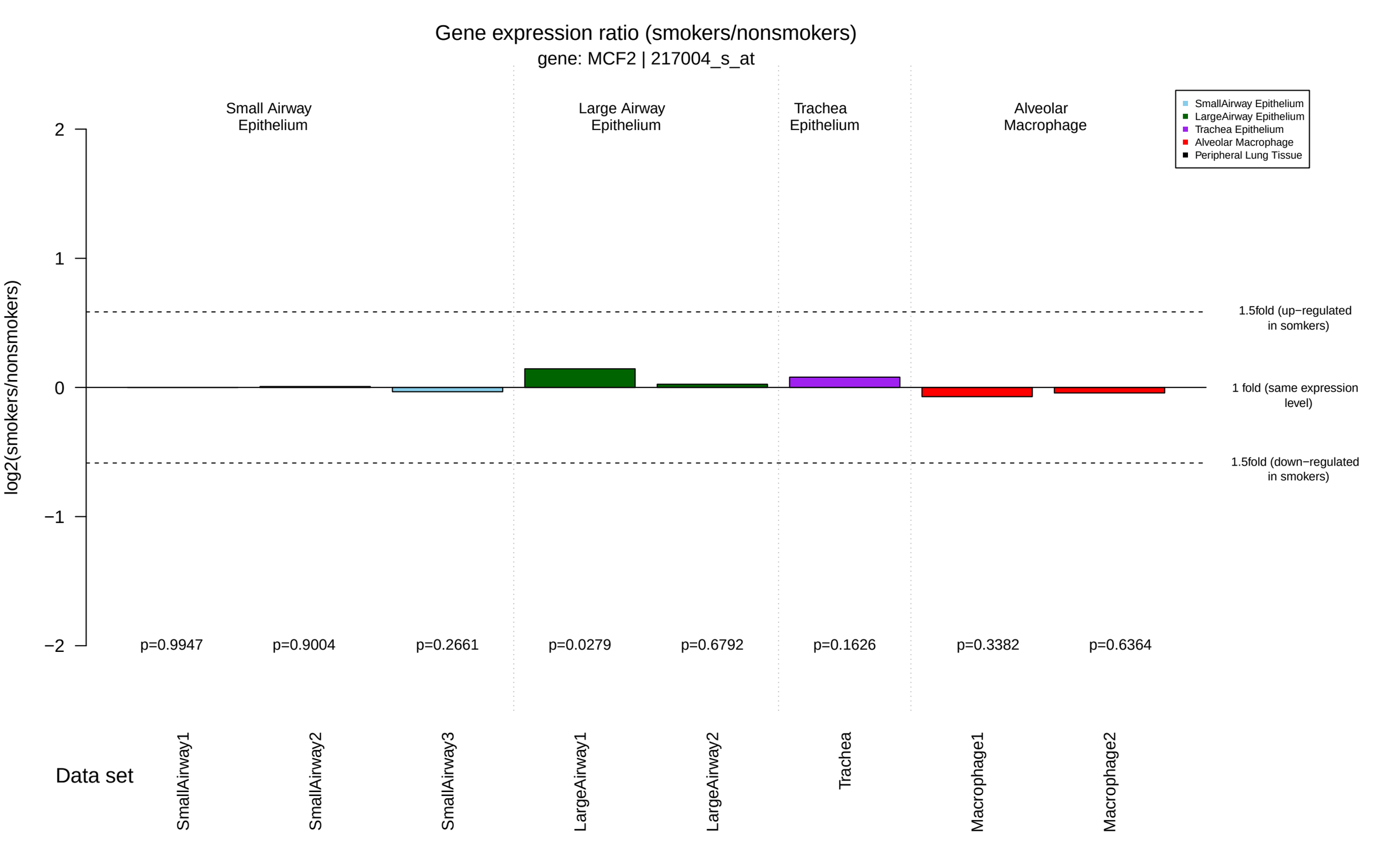
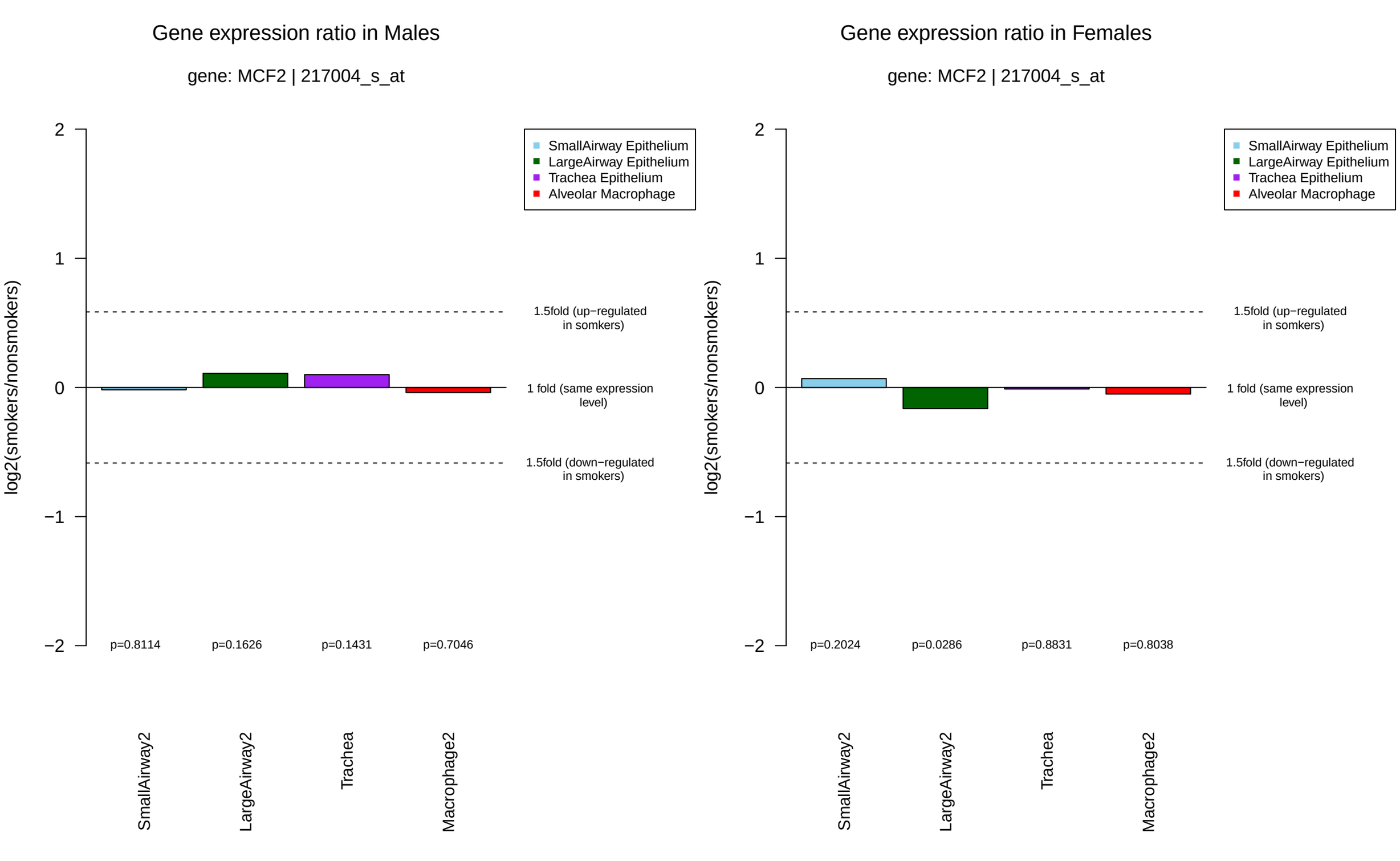
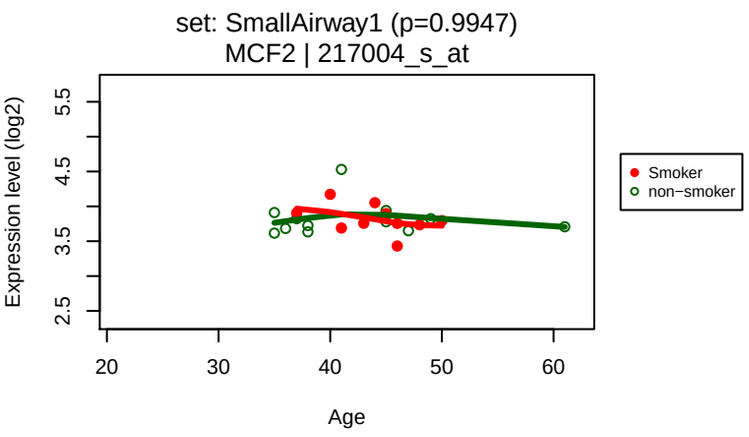
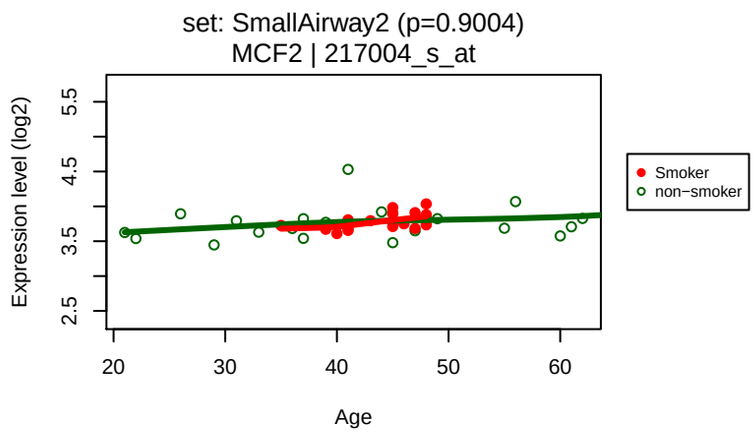
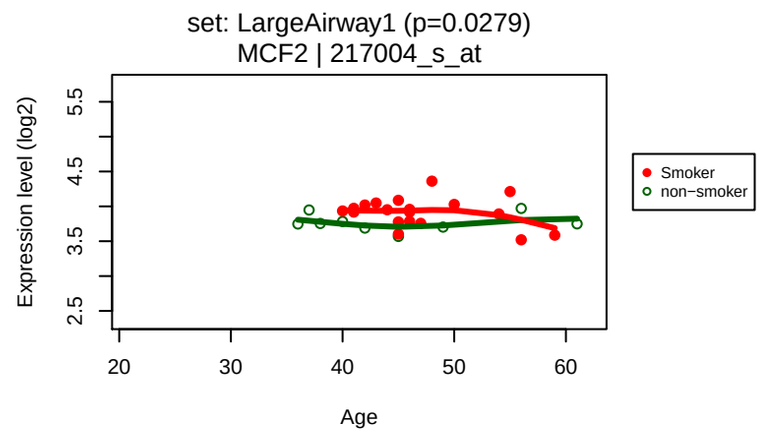

Gene expression of smokers and nonsmokers
gene: MCF2 | 217004_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9947
p=0.9004
p=0.2661
p=0.0279
p=0.6792
p=0.1626
p=0.3382
p=0.6364
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

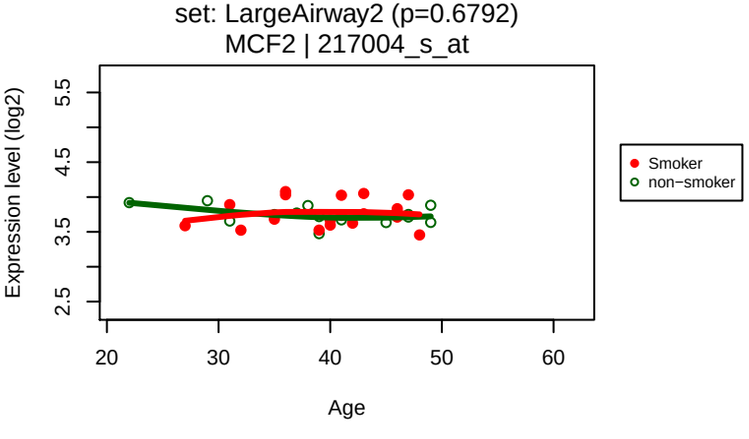

Gene expression ratio (smokers/nonsmokers)
gene: MCF2 | 217004_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9947
p=0.9004
p=0.2661
p=0.0279
p=0.6792
p=0.1626
p=0.3382
p=0.6364
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

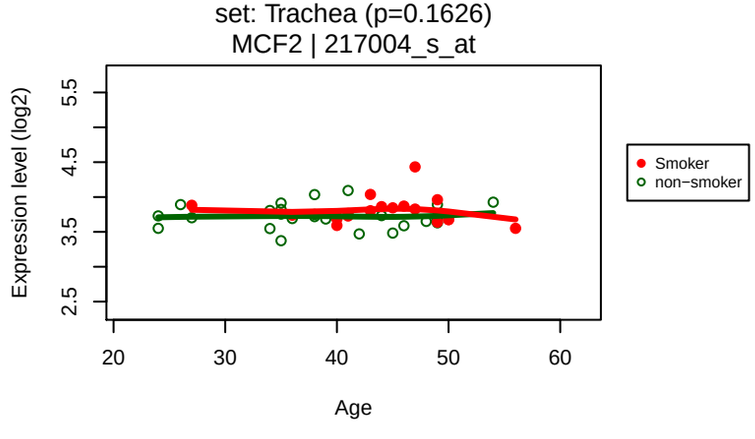

Gene expression ratio in Males
gene: MCF2 | 217004_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8114
p=0.1626
p=0.1431
p=0.7046
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCF2 | 217004_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2024
p=0.0286
p=0.8831
p=0.8038
SmallAirway2
LargeAirway2
Trachea
Macrophage2

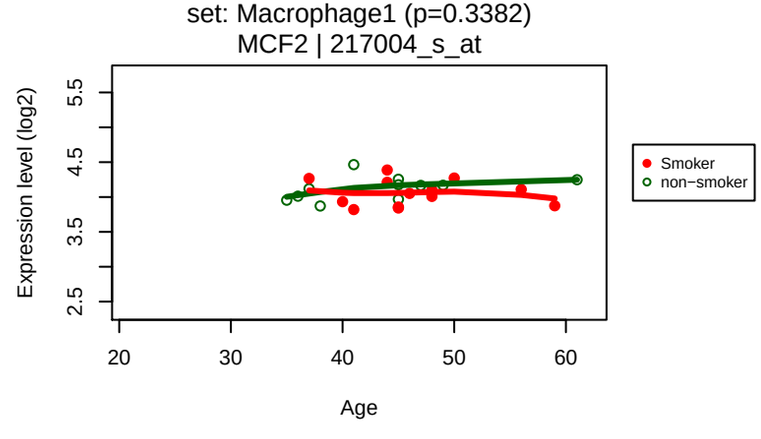

set: SmallAirway1 (p=0.9947)
MCF2 | 217004_s_at
Expression level (log2)
Age
Smoker
non−smoker

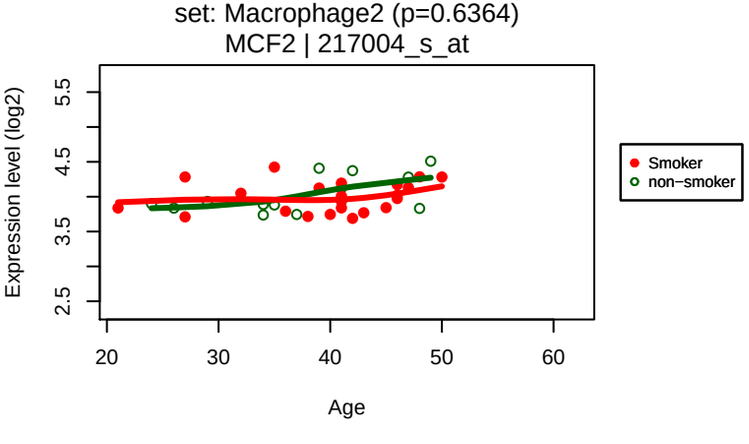

set: SmallAirway2 (p=0.9004)
MCF2 | 217004_s_at
Expression level (log2)
Age
Smoker
non−smoker

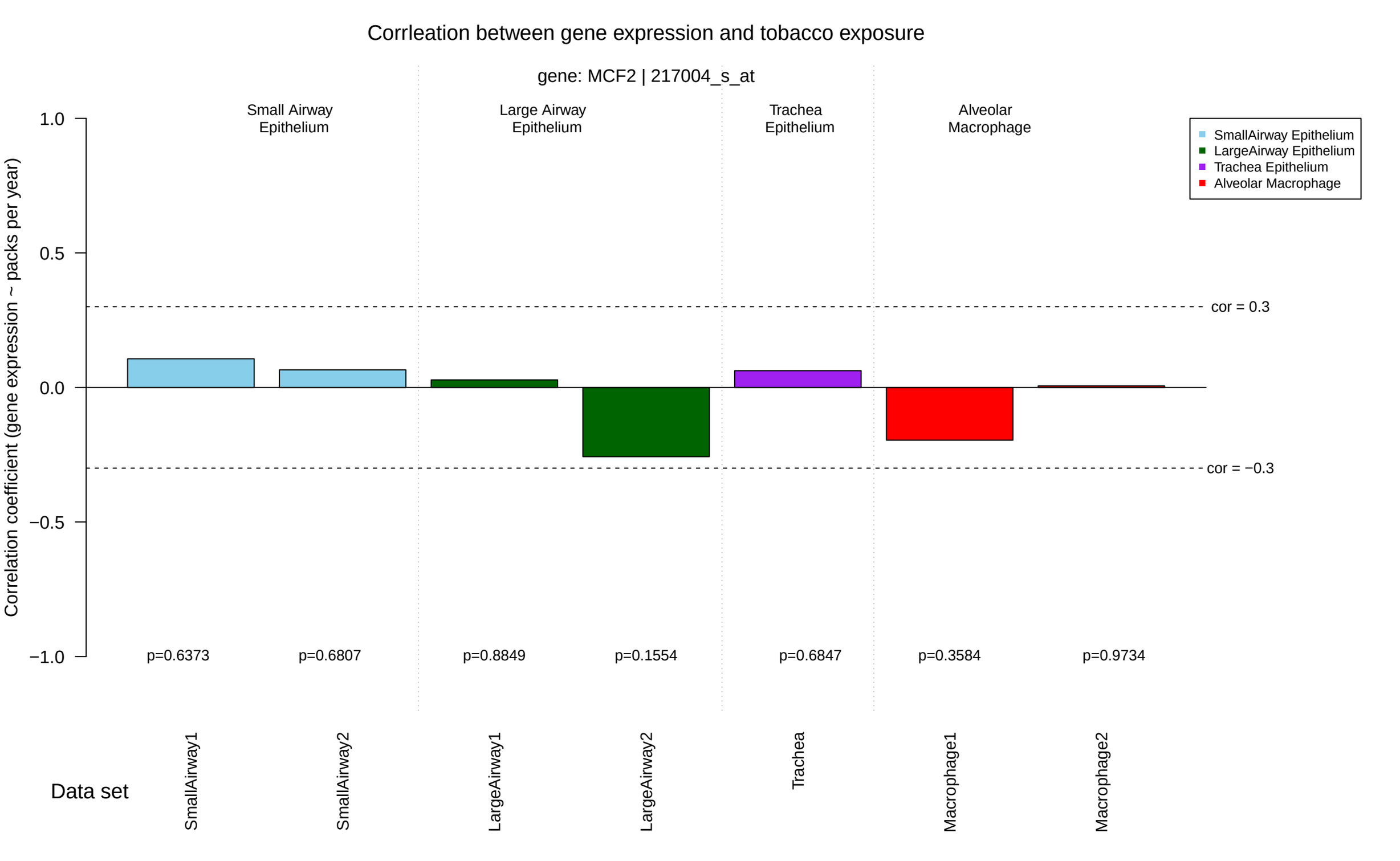

set: LargeAirway1 (p=0.0279)
MCF2 | 217004_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6792)
MCF2 | 217004_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1626)
MCF2 | 217004_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3382)
MCF2 | 217004_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6364)
MCF2 | 217004_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCF2 | 217004_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6373
p=0.6807
p=0.8849
p=0.1554
p=0.6847
p=0.3584
p=0.9734
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2